

Multi-Agent Systems (MAS) in Financial Engineering

Architecture, Autonomous Orchestration, and Financial Applications

Prof. Chae Shick Chung

Graduate School of Economics / MBA
Sogang University

AI Finance Seminar

Executive Summary & Learning Objectives

From Passive LLMs to Multi-Agent Systems (MAS)

Modern AI in finance has evolved beyond simple chat interfaces and passive RAG. **Multi-Agent Systems (MAS)** deploy specialized, autonomous agents that collaborate, debate, and programmatically orchestrate complex financial workflows.

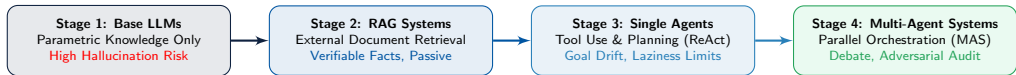
Key Architectural Innovations

- **Specialized Personas:** Division of cognitive labor.
- **Consensus & Debate:** Reducing individual agent bias.
- **Programmatic Orchestration:** Dynamic sub-agent spawning via CLIs (Claude Code, Codex CLI).

Target Financial Capabilities

- **M&A Due Diligence:** Footnote auditing & adjustment.
- **Asset Allocation:** Multi-persona consensus voting.
- **Live Market Defense:** Streaming lineage & noise filtering.
- **Quant Execution:** Autonomous sandbox coding.

Evolution of Financial AI Architectures



Why Single Agents Fail in Complex Financial Workflows

- **Agentic Laziness:** Skipping items in 100-page SEC filings after inspecting the first 20.
- **Self-Preferential Bias:** Inability to critically audit its own DCF or valuation assumptions.
- **Goal Drift:** Loss of negative constraints (e.g., ESG / risk limits) over long multi-turn sessions.

1. Hermes Agents: Multi-Stage Task Decomposition

First Screen / Initial Interface UI:

```
● ● ● Hermes Financial Research Agent v2.4
-----
$ hermes analyze --target "Acme Corp"
--mode audit
[SYSTEM] Initializing 3-Stage Pipeline...
[STAGE 1: SEARCH] Ingesting SEC 10-K
Footnotes...
[STAGE 2: QUANT] Recalculating Adjusted
EBITDA...
[STAGE 3: AUDIT] Cross-referencing
litigation note...
-----
[HERMES STATUS] Fact Lineage Anchored |
Audit OK
```

Core Architecture

Runs a rigid 3-stage pipeline: **Search** → **Analysis** → **Validation**. Mismatches in Stage 3 force reflection back to Stage 2.

Financial Application: M&A Footnote Audit

- **Scenario:** Target firm reports a 30% spike in R&D expenses.
- **Hermes Execution:**
 - 1 **Search Agent:** Ingests 10-K Footnote 14 (Litigation).
 - 2 **Analysis Agent:** Reclassifies \$40M patent settlement buried in R&D.
 - 3 **Validation Agent:** Confirms true operational R&D decreased by 12%.
- **Outcome:** Prevents overpaying for M&A target based on inflated R&D metrics.

2. Buzz (Buzz Block): Multi-Agent Consensus Voting

First Screen / Initial Interface UI:

```
● ● ● Buzz Block Framework --- Consensus
Board
-----
[MACRO AGENT]: "Fed rate hike (+50bps) ->
Cut Tech."
[EQUITY AGENT]: "Semiconductor margins
strong."
[RISK MANAGER]: "Sharpe ratio drops below
1.2 limit."
--- CONSENSUS PROTOCOL: Weighted Majority
Voting ---
[FINAL DECISION]: Rebalance: Tech 35% ->
25%, Cash 25%
```

Core Architecture

Assigns distinct financial personas with private tools.
Agents interact over a shared **Message Bus** using
consensus voting.

Financial Application: Macro Shock Rebalancing

- **Scenario:** Sudden FOMC interest rate surprise shock.
- **Buzz Execution:**
 - 1 Macro Economist: Models yield curve steepening.
 - 2 Equity Analyst: Evaluates company-level debt ratios.
 - 3 Portfolio Manager: Re-calculates Mean-Variance weights.
- **Outcome:** Mitigates single-model bias through collective multi-persona judgment.

3. GrokBot: Real-Time Streaming & Dynamic Lineage

First Screen / Initial Interface UI:

● ● ● GrokBot Real-Time Market Terminal

```
[STREAM] 14:02:11 -- FX Spike: USD/KRW  
1,380  
[LINEAGE TRACER]: Scanning source  
anchors...  
[ALERT]: Unverified Twitter rumor on  
capital controls.  
[VERIFIED]: Bank of Korea Press Release  
#402  
[ACTION]: Signal filtered as FALSE NOISE.
```

Core Architecture

Directly interfaces with ultra-low latency streaming APIs and attaches **Dynamic Provenance Anchors**.

Financial Application: Flash Crash Defense

- **Scenario:** Fake social media rumor triggers FX volatility.
- **GrokBot Execution:**
 - 1 Captures streaming news feed in milliseconds.
 - 2 Checks source lineage against official Central Bank feed.
 - 3 Flags social media rumor as unanchored noise.
- **Outcome:** Protects algorithmic trading desks from false panic-selling.

4. OpenAI Dot / Codex CLI: Autonomous Code Sandbox

First Screen / Initial Interface UI:

```
● ● ● OpenAI Dot / Codex CLI Sandbox
-----
$ codex-cli run --script backtest.capm.py
[EXEC] Running Python 3.12 in isolated
sandbox...
[ERROR] KeyError: 'Adj Close' in
DataFrame.
[SELF-CORRECT]: Patching yfinance
multi-index...
[SUCCESS] Backtest complete. Sharpe: 1.84
```

Core Architecture

A CLI-first agent that writes, executes, tests, and self-corrects Python/C++ quantitative scripts inside a secure local sandbox.

Financial Application: Quant Backtesting

- **Scenario:** Analyst requests backtesting Fama-French 3-factor strategy on KOSPI stocks.
- **Codex Execution:**
 - 1 Fetches ticker data via yfinance.
 - 2 Writes regression script and runs in sandbox.
 - 3 Autonomously fixes data indexing errors.
- **Outcome:** Generates verified backtest report without human coding overhead.

Programmatic Parallel Orchestration: Claude Code & Codex CLI

From Static Agent Chains to Dynamic JS/CLI Workflows

Instead of relying on a fixed agent graph, the orchestrator agent **writes its own custom orchestration script on the fly** using special primitives:

Core Orchestration Primitives

- `agent()`: Spawns isolated sub-agents with dedicated models (sonnet, haiku).
- `parallel()`: Executes dozens/hundreds of sub-agents in parallel (Fan-out).
- `pipeline()`: Streams outputs sequentially through verification stages.

Key Architectural Benefits

- **Context Isolation**: Prevents context window bloating and performance degradation.
- **Checkpointing**: Resume long-running audits from saved state.
- **Cost Optimization**: Assigns cheap models (haiku) to routine sub-tasks.

6 Dynamic Workflow Patterns in Financial Engineering

1. Classify-and-Act

Routes incoming high-risk signals to quarantined sub-agents before main execution.

2. Fan-out and Synthesize

Spawns 500 sub-agents to parse S&P 500 10-K covenant clauses simultaneously.

3. Adversarial Verification

Pairs a generation agent with a dedicated skeptic agent to audit accounting entries.

4. Generate-and-Filter

Generates 200 candidate basket stocks, filtering by strict ESG and liquidity rules.

5. Tournament (Pairwise)

Runs competing valuation models (DCF, LBO, Comps) and selects the most robust model.

6. Loop until Done

Iterates trade reconciliation until all discrepancy line items reach zero.

Comparative Matrix of Multi-Agent Frameworks

Framework	Core Architecture	Latency	Cost / Token	Best Financial Fit
Hermes Agents	3-Stage Pipeline (Search/Quant/Audit)	Medium	Moderate	M&A Due Diligence & Footnotes
Buzz Block	Persona Consensus (Message Bus)	Medium-High	Moderate-High	Asset Allocation & Rebalancing
GrokBot	Real-Time Feed + Lineage Anchors	Low (< 1s)	Low	Live Market News & Crash Defense
OpenAI Dot / Codex	Sandbox Code Execution	Medium	Low-Moderate	Quant Scripting & Backtesting
Claude Dynamic Workflows	Programmatic Parallel JS Spawning	Scalable	Dynamic	Portfolio-Wide 10-K Audits

Strategic Integration Rule for Financial CIOs

Use **GrokBot** for low-latency market feeds, **Codex CLI** for internal quantitative execution, and **Claude Dynamic Workflows** for large-scale document audits.

1. Systemic Risk: Correlated Panic

If thousands of autonomous trading agents share similar base LLM weights and loss-aversion thresholds, a macro shock can trigger **correlated, simultaneous panic-selling**, destroying market liquidity.

2. Quarantine & Injection Defense

Untrusted external feeds (social media, untrusted PDFs) must be processed in **quarantine sub-agents** with zero system execution privileges to block prompt injection attacks.

3. Fiduciary Duty & Sign-Off

AI agents cannot hold legal personhood. The financial analyst remains 100% legally accountable for all published reports and trade executions.

Human-in-the-Loop (HITL) Guardrails

- **Traceability Check:** Audit numerical citations.
- **Caveat Restoration:** Re-insert omitted risks.
- **Final Approval:** Human signature required.

Seminar Discussion & Hands-on Assignment

Seminar Discussion Topic

”Should financial institutions rely on proprietary cloud-orchestrated MAS (e.g., Anthropic/OpenAI) or build local, open-source multi-agent networks (e.g., Hugging Face/Ollama) to preserve data sovereignty?”

Hands-on Student Assignment

- 1 Open the `module_4_practice_v3.py` notebook.
- 2 Configure a 3-agent consensus team (Macro, Equity, Risk Manager) using the **Buzz** voting protocol.
- 3 Test portfolio rebalancing under a simulated +100bps interest rate shock and evaluate Sharpe ratio stability.